

Abstract - Large Language Models have significantly improved the ability of artificial intelligence systems to engage in natural and context-aware conversations. However, maintaining relevant information across long interactions remains a major challenge due to limited context windows, increasing memory requirements, and rising computational costs. As conversations grow, important details may be forgotten while redundant information consumes valuable processing resources. This paper presents a smart memory compression framework designed to enhance long-term conversational efficiency by identifying, prioritizing, and retaining important memories while compressing or removing less relevant content. The proposed approach integrates memory ranking, semantic similarity analysis, and summarization techniques to optimize the information passed to the language model. By preserving critical contextual knowledge and reducing unnecessary data, the framework aims to improve response quality, conversational continuity, and system scalability. The methodology supports efficient token utilization while maintaining contextual understanding during extended interactions. This research contributes toward the development of intelligent conversational agents capable of handling long-term dialogues more effectively, enabling more personalized, consistent, and resource-efficient AI systems for real-world applications.

Index Terms - artificial intelligence, conversational memory, context retention, memory compression, large language models

1. INTRODUCTION

Artificial Intelligence (AI) has transformed the way humans interact with digital systems by enabling machines to understand, process, and generate human-like language. Recent advancements in Large Language Models (LLMs) such as GPT, Llama, and Mistral have significantly improved conversational intelligence, allowing AI systems to provide context-aware responses across a wide range of applications. These models are increasingly used in virtual assistants, customer support systems, educational platforms, and enterprise solutions where maintaining meaningful conversations over extended periods is essential.

Despite their remarkable capabilities, LLMs face significant challenges when handling long conversations. Most models operate within a limited context window, restricting the amount of information that can be processed at a given time. As conversations grow longer, earlier interactions may be forgotten, resulting in loss of important context and reduced response consistency. Furthermore, storing and processing the entire conversation history increases computational complexity, memory consumption, latency, and operational costs. To address these challenges, researchers have explored various memory management techniques, including external memory modules, retrieval-based systems, and conversational summarization methods. However, many existing approaches either fail to preserve critical contextual information or introduce excessive computational overhead. This paper proposes a Smart Memory Compression Framework for Long AI Conversations. The framework identifies and prioritizes important conversational memories, compresses redundant or low-priority information, and retains essential context required for generating accurate and coherent responses. By combining memory ranking, semantic similarity analysis, and intelligent summarization, the proposed approach aims to optimize token usage while maintaining conversational continuity and response quality. The framework contributes toward building scalable, efficient, and personalized AI systems capable of supporting long-term human-AI interactions.

2. LITERATURE REVIEW

The challenge of maintaining contextual understanding in long AI conversations has attracted significant attention in recent years. As Large Language Models (LLMs) become increasingly capable of handling complex interactions, researchers have focused on developing efficient memory mechanisms that preserve important information while minimizing computational overhead. Chen et al. introduced the COMEDY framework in their work “Compress to Impress: Unleashing the Potential of Compressive Memory in Real-World Long-Term Conversations.” The study addresses the limitations of traditional memory systems in handling lengthy conversations by proposing compressive memory architecture. The framework selectively compresses historical conversation data while preserving critical contextual information. Experimental results demonstrated that the proposed approach improves long-term dialogue consistency and enhances the ability of conversational agents to recall relevant information from earlier interactions. To further

improve long-context understanding, Yue et al. proposed Long Mem: Augmenting Language Models with Long-Term Memory. Their approach extends the capabilities of language models through the integration of an external memory module that stores and retrieves historical information when required. The study showed that separating memory storage from the model's fixed context window significantly improves performance in tasks requiring long-range reasoning and contextual recall. Similarly, Wang et al. presented Memory Bank: Enhancing Large Language Models with Long-Term Memory, which introduces a structured memory storage and retrieval framework for persistent conversations. The proposed system organizes conversation history into memory units that can be efficiently retrieved based on relevance. This approach improves conversational continuity and allows AI systems to maintain personalized interactions over extended periods. In a subsequent study, Renyuan Wang et al. explored additional techniques for enhancing large language models with long-term memory capabilities. Their research focused on memory retrieval strategies that enable models to access previously stored information more effectively. The findings highlighted the importance of intelligent memory selection in improving response consistency and reducing information loss during lengthy interactions. While the previous studies primarily focused on conversational memory, Han et al. addressed memory efficiency from a neural network perspective through their influential work "Deep Compression: Compressing Deep Neural Networks with Pruning, Trained Quantization and Huffman Coding." The authors demonstrated that neural networks can be significantly compressed without substantial loss in accuracy. Their compression techniques reduced storage requirements and inspired later research on efficient memory utilization in AI systems. Another important contribution was made by Kwon et al. in "Efficient Memory Management for Large Language Model Serving with Paged Attention." The authors introduced the Paged Attention mechanism to address memory fragmentation and improve resource utilization during LLM deployment. Their approach enhanced serving efficiency and reduced memory wastage, making large-scale language model applications more practical and cost-effective. Shan et al. examined memory architectures in their work "Cognitive Memory in Large Language Models." Inspired by human cognitive processes, the study analyzed sensory memory, short-term memory, and long-term memory mechanisms within LLMs. The authors discussed how memory compression and retrieval strategies can improve contextual understanding while maintaining computational

efficiency. Their findings provide valuable insights into designing more human-like memory systems for conversational AI. More recently, Zhang et al. proposed the "Experience Compression Spectrum: Unifying Memory, Skills, and Rules in LLM Agents." The study introduced a unified framework that compresses experiences, learned skills, and operational rules into compact representations. This approach enables AI agents to manage long-term experiences more efficiently while supporting complex reasoning and decision-making tasks. The research highlights the growing importance of adaptive memory compression techniques for future intelligent systems. Although existing studies have made substantial progress in long-term memory management, challenges remain in balancing memory retention, contextual relevance, computational efficiency, and response quality. Most approaches either focus on memory retrieval or compression independently. Therefore, there is a need for an integrated framework that combines memory ranking, semantic understanding, and intelligent compression to optimize long conversational interactions. The proposed Smart Memory Compression Framework aims to address these limitations by retaining critical information while reducing redundant conversational content, thereby improving the efficiency and effectiveness of long-term AI conversations.

3. METHODOLOGY

The Smart Memory Compression Framework is designed to improve the efficiency of Large Language Models when handling long and complex conversations. The system focuses on preserving important contextual information while reducing the amount of redundant data processed by the model. By intelligently managing conversational memory, the framework aims to maintain response quality while minimizing memory consumption and token usage. The architecture consists of six major components that work together to process, analyze, compress, and retrieve conversational information. The process begins when a user submits a message to the system. The incoming conversation is first analyzed to determine the relevance and importance of different pieces of information. Important conversational details are retained, while repetitive or low-priority content is identified for compression. A Memory Ranking Module is used to assign priority scores to conversation segments based on their contextual significance. The ranked information is then passed to the Semantic Similarity Module, which detects duplicate or closely related content using semantic analysis techniques. This helps prevent unnecessary

storage of repetitive information. The Compression and Summarization Module generates concise representations of less important conversation segments while preserving key information required for future interactions. The compressed data, along with highly relevant memories, is stored in a dedicated long-term memory repository. Before generating a response, the Context Optimization Module retrieves the most relevant memories from the repository and combines them with the current user query. This optimized context is then supplied to the Large Language Model for response generation. Finally, the Response Generation Module produces context-aware and coherent responses while utilizing significantly fewer tokens than traditional approaches. The proposed architecture enables efficient memory utilization, improved contextual retention, reduced computational overhead, and enhanced conversational continuity, making it suitable for long-term AI interactions.

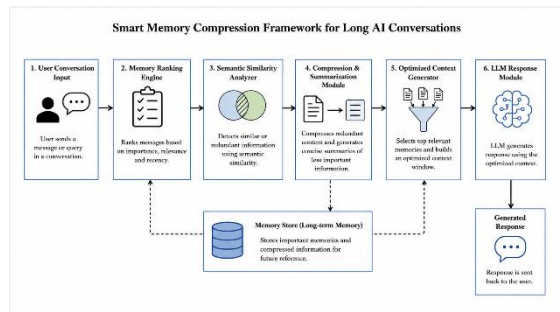


Figure 1. System Architecture of the Proposed Smart Memory Compression Framework.

The Memory Ranking Module is responsible for determining the importance of information within a conversation. During long interactions, not every message contributes equally to the overall context. Some messages contain critical details, preferences, or decisions, while others may be temporary or repetitive. This module analyzes conversational content and assigns a priority score based on factors such as relevance, frequency of occurrence, and contextual significance. Information with higher scores is considered valuable and retained for future use, whereas low-priority content is marked for compression or removal. By filtering information according to its importance, the module helps reduce memory overhead while ensuring that essential conversational details remain available for generating accurate and context-aware responses. The Semantic Similarity Module is designed to identify messages that convey similar meanings within a conversation. In long interactions, users often repeat information in

different ways, leading to unnecessary duplication in memory storage. To address this issue, the module evaluates the semantic relationship between conversation segments using natural language processing techniques. Messages that share similar intent or meaning are grouped together, allowing redundant information to be minimized. This process helps optimize memory utilization without affecting the overall understanding of the conversation. By focusing on the meaning of the text rather than exact word matching, the module contributes to more efficient memory management and supports the retention of unique and meaningful information. The Compression and Summarization Module focuses on reducing the size of stored conversational data while preserving important information. Instead of retaining every message in its original form, the module generates concise summaries of low-priority or repetitive content. The summarization process captures the key ideas and contextual details required for future reference while eliminating unnecessary information. This approach significantly reduces memory consumption and token usage without compromising the quality of responses. By creating compact representations of past interactions, the module enables the system to maintain long-term conversational context in a more efficient manner and supports scalable deployment of AI-based conversational systems. The Long-Term Memory Repository serves as a centralized storage component for important conversational information. It maintains both high-priority memories and compressed summaries generated during previous interactions. The repository is designed to organize and store contextual information in a structured manner, allowing efficient retrieval whenever required. Unlike traditional conversation histories that store every message sequentially, the repository focuses on preserving meaningful information that contributes to future conversations. This helps reduce storage requirements while ensuring continuity across multiple interactions. By maintaining a persistent memory structure, the system can provide more personalized, consistent, and context-aware responses over extended periods of communication. The Context Optimization Module is responsible for preparing the most relevant information before it is sent to the Large Language Model. When a user submits a query, the module retrieves important memories from the repository and evaluates their relevance to the current conversation. Only the most useful contextual information is selected and combined with the user's latest input. This process prevents the model from processing unnecessary data and helps maintain an efficient context window. By delivering a refined and optimized set of information, the module improves

response quality, reduces computational cost, and enables the language model to generate more accurate and contextually relevant outputs. The Response Generation Module represents the final stage of the proposed framework. After receiving the optimized conversational context, the Large Language Model processes the information and generates an appropriate response for the user. Since irrelevant and redundant content has already been filtered out, the model can focus on the most important aspects of the conversation. This improves response consistency, contextual accuracy, and overall conversational quality. Additionally, the reduced context size lowers token consumption and processing time, making the system more efficient. The generated response is then delivered to the user, completing the conversational cycle while preserving important information for future interactions.